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Feature Fusion and Explainable Deep Learning Framework for Intelligent Skin Disease Classification Using Clinical Dermatology Images

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Abstract
Skin diseases represent a significant burden in healthcare management due to delayed diagnosis, visual complexity, and variability in clinical decision-making. Manual methods are prone to misdiagnosis, time-consuming, and often lack consistency. With the rise of clinical dermatology cases, there is a critical need for automated, explainable systems to support reliable skin disease diagnosis. This study proposes an Explainable Artificial Intelligence (XAI)-driven Hybrid Multi-Class Skin Disease Classification (HMSC) framework designed for a clinical dermatology system. This research work utilized benchmark Dermnet dataset for the evaluation of HMSC framework. The framework integrates deep learning models VGG16, EfficientNet-B0, Inception-V3, and MobileNet-V3 for automatic feature extraction and fusion from dermoscopic images of five common skin conditions: acne, psoriasis, eczema, urticaria, and scabies. LIME XAI technique is incorporated to improve interpretability by visually highlighting the most influential image regions in each prediction. The HMSC proposed framework gives an accuracy of 98.70%, with 98% precision, 98.70% recall, 98.50% specificity, 99.30% AUC-ROC, 96% MCC, and 98.30% F1-score. The model exhibited excellent cross-class generalization and interpretability of skin diseases on test data. The reliability and stability of the proposed framework are further tested using k-fold cross-validation technique and testing it on another 2 independent datasets, HAM1000 and ISIC-2019. These results suggest that the HMSC framework is a reliable and transparent diagnostic framework for skin disease classification. HMSC scalability and explainability show the potential for its real-world deployment in healthcare systems after some ethical and diverse demographic skin tone considerations. This framework has the capacity to minimize the clinician workload with improved diagnostic efficiency and support consistent decision-making.

Keywords Skin disease classification · Explainable artificial intelligence (XAI) · Clinical dermatology images · Deep learning · Medical image analysis · Feature fusion · Multi-class classification



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1 Introduction

Artificial Intelligence (AI) is rapidly reshaping digital healthcare by enabling consistent, faster, and accurate diagnostic support across diverse medical imaging domains. Skin diseases are highly prevalent worldwide, affecting quality of life through psychological distress, physical discomfort, and financial burden. Manual assessment of dermatological conditions is prone to misdiagnosis and even time-consuming [1]. AI-based decision support systems, particularly those enhanced with explainable techniques, can assist clinicians by providing reliable second opinions and improving diagnostic confidence.

Although substantial research has been conducted in this domain, practitioners still require reliable and explainable diagnostic solutions for major diseases such as hepatitis [2], cancer [3], and diabetes [4]. Skin is the largest body organ, acting as a barrier to the body's internal organs. It is most prone to infections due to its interaction with the environment [5]. Disorders like psoriasis and eczema may be uncomfortable, cause chronic sleeping difficulties and itching, as well as hideous symptoms of redness and scaling, that affect self-esteem and self-confidence negatively [6]. These conditions often lead to anxiety, chronic depression, and stress, affecting both lifestyle quality and emotional stability [7]. Skin disease and cancer treatment are very expensive and put a lot of financial burden if treated at a later stage [8].

Traditional diagnostic approaches, including physical examination, imaging, and histopathology, differ from one region to another. Some modern methods (e.g., laboratory tests and biopsies) are widely used, but these methods' results are totally dependent on human interpretation, as well as they are labor-intensive and time-consuming [9, 10]. Therefore, the development of an AI-based skin disease and cancer types diagnostics framework is the need of this time, where everyone is exposed to multiple infections and harmful radiation.

Deep learning models, especially Convolutional Neural Networks (CNNs), have demonstrated excellent performance as a single baseline model in medical image categorization by automatically learning spatial feature hierarchies [11]. Deep learning–based skin cancer classification has gained substantial traction, with recent works exploring enhanced preprocessing, hybrid feature extractors, and meta-heuristic optimization to improve diagnostic accuracy [12]. The DSC-EDLMGWO model advances this direction by integrating SE-DenseNet features with an optimized ensemble of LSTM, ELM, and SSDA, demonstrating strong performance of accuracy value 98.38% and 98.17% on HAM10000 and ISIC datasets. Another approach [13] combining preprocessing (noise/hair removal), optimized segmentation techniques such as Region Growing with meta-heuristics, and lightweight CNN architectures have shown strong improvements in lesion localization and classification accuracy and precision of 99.01% and 98.6%. The reviewed work aligns with this trend by integrating MHBO-enhanced segmentation with MobileSkinNetV2, demonstrating competitive performance on ISIC datasets. In dermatology, various machine vision techniques (MVT) and CNN architectures have been applied for computer-aided diagnosis (CAD) of skin disorders [14–16].

The study is organized as follows: in Sect. 2, we give a summary of recent research on this issue. Section 3 introduces the dataset and deep learning methods utilized for skin cancer prediction. It also includes the proposed technique and provides a thorough description of the approach. The results are discussed in Sect. 4. Finally, in Sect. 5, the conclusion of this research is made.

2 Related Work

Over the years, considerable research work, which has consisted of decades, has been conducted in this area of skin diseases. This section provides a brief overview of some recent and advanced methodologies. Table 1 shows the literature review matrix.

In [17], CNN-based models ResNet and VGG16 are used for skin cancer detection and classification. Despite high accuracy, their research still lacks in terms of interpretability and the class-imbalance problem, which is a major concern in real-world settings and deployment. They named their model DCAN-Net by integrating a

Table 1 Literature review table

Refs.	Year	Objective	Methodology	Results	Conclusion
[17]	2025	Develop a CNN-based framework for accurate skin cancer detection	DCAN-Net model with Grad-CAM explainability; HAM10000 dataset	Precision: 97.00%, Recall: 97.57%, Accuracy: 97.57%	Explainable AI enhances clinician trust and improves skin cancer detection
[18]	2024	Improve deep learning-based skin disease diagnosis accuracy	CNN-based model with ISIC dataset; tested shallow vs. deep learning models	CNN achieved 87.64% accuracy on three disease categories	Deep learning can enhance early-stage skin disease diagnosis
[19]	2024	Develop a multi-model deep learning system for diagnosing multiple skin conditions	Xception model with transfer learning; classified 40 skin diseases	Accuracy 95%	Multi-model deep learning surpasses traditional methods in skin disease diagnosis
[20]	2023	Detect skin diseases using CNN-based image processing	Image processing, segmentation, classification using CNN	Fast, simple, and effective classification	AI-assisted image processing can reduce diagnostic errors
[21]	2023	Develop an automated CNN-based dermatological diagnosis system	CNN with Softmax classifier; preprocessing and feature extraction	Improved accuracy over traditional methods; real-time classification potential	AI-driven classification enhances dermatological education and diagnostics
[22]	2023	Develop a CNN-based model (MPXV-CNN) for classifying mpox lesions	Trained on image data set; tested sensitivity, specificity, AUC across datasets	Sensitivity: 0.83−0.91, Specificity: 0.89−0.96, AUC: 0.966	AI can aid in early mpox detection across skin tones and body regions
[23]	2023	Develop a CNN model for early detection and classification of six skin diseases	Preprocessing, feature extraction, CNN-based classification	Accuracy 98%; significant improvement in early-stage detection	AI can support dermatologists by providing high-accuracy early diagnosis
[24]	2022	Improve skin disease classification using AI and spectral centroid magnitude (SCM)	Hybrid approach: CNN with enhanced localization and spectral features	Higher accuracy compared to traditional methods; better feature extraction	Hybrid AI methods improve skin disease classification performance
[25]	2021	Develop a lightweight skin disease classification system for mobile applications	MobileNet V2 + LSTM; HAM10000 dataset; compared with CNN, VGG	Achieved 85% accuracy, lower computation cost, and better efficiency on mobile devices	The model can assist practitioners in early skin disease detection
[26]	2021	Evaluate CNN models for diagnosing vitiligo and compare their accuracy with human raters	Trained CNN models on Chinese in-house and public datasets; compared results with dermatologists	CNN models outperformed intermediate raters and matched expert-level diagnosis	CNNs can diagnose vitiligo effectively without Wood's lamp images
[27]	2018	Develop an automated system for skin disease recognition using CNN	Image filtering, feature extraction using CNN, classification via logistic regression	Faster, more accurate diagnosis than traditional methods	CNN-based system enhances dermatological diagnostics and education
[28]	2022	Develop a mobile-based application for diagnosing skin diseases using deep learning	Image preprocessing (OpenCV), feature extraction, classification using ANN	Achieved reliable detection of Nevus and Ringworms with minimal errors	AI-based image classification can provide affordable dermatological screening

ConvNeXt-based CNN with an enhanced attention mechanism and Grad-CAM for enhanced feature extraction and explainability on the HAM10000 dataset.

In [19], researchers move in the reverse direction and give a solution to diagnose skin disease using machine learning approaches such as KNN and SVM instead of deep learning models. They target fewer types of skin diseases and, therefore, this approach lacks generalization across diverse skin diseases. This approach, although it provides high accuracy but cannot be used in the real world.

Advances in deep learning have made automated classification of skin disorders much more accurate; however, concerns over trust, interpretability, and generalizability remain. As an instance, an XAI-based system has been suggested to augment clinical trust by offering clarity into the understanding of the diagnostic decision-making process. Another paper used ISIC 2019 and added LIME explanations to a deep learning model, which reached 94.47% accuracy, locating eight classes of the skin lesions correctly and providing visual feedback to clinicians so that they could verify the model [29]. In a similar case, SkinSage XAI developed a large-scale, customized HAM10000 dataset and applied Inception-V3, which could be explained using Grad-CAM and LIME visualizations, and shows high-accuracy rates of 96% and AUC of 99.83%, demonstrating the advantage of both performance and interpretability employing such methods [30]. ResNet50V2, VGG16, and InceptionResNetV2 have been further explained using ResNet50V2 using XAI techniques like LIME as a means of transparency, supporting the argument that interpretability should be taken seriously in achieving clinical acceptability of AI-powered dermatology applications [31].

In addition to the conventional CNN-based systems, researchers examined transfer learning and transformers in skin disease detection. Layer-wise relevance propagation (LRP) was applied to discover the symptom-related regions, a transfer learning approach with pre-trained VGG16 showed 93.29% accuracy in classifying the virus skin disease sample, including chickenpox, measles, and monkeypox[32]. Most recently, DinoV2 and Vision Transformers (ViTs) have surpassed CNN records, with a test accuracy of 96.48% and an F1-score of 97.27% on a 31-class dataset, but still retain their explanability with Grad-CAM and SHAP explanations [33]. Overall, these studies show that deep learning with interpretability is the future of clinical dermatology that can help make a significant improvement in the reliability of diagnosing and lead to trust of the clinical dermatology system.

In [20], researchers presented research that primarily focuses on detecting skin diseases using image-processing techniques. It involves a simple procedure of capturing an image of a blown-up area, after which the analysis is performed using image processing techniques. The result determines the particular skin condition. The possibility of real-time detection is offered through the employment of CNN, which contributes to the considerable acceleration and enhancement of diagnosis based on the processed images without compromising their accuracy.

Another is focused on the precise classification, detection, and as comprehensive as possible information on skin diseases. The steps include several steps; initial images of skin diseases undergo preprocessing, which is then followed by feature extraction. The processed images are then judged in a few steps with a Deep Convolutional Neural Network (DCNN). This expected strategy is imminent due to its ease, expediency, and high accuracy, having reached 98.01% accuracy [23]. In [24], researchers described the intense role of artificial intelligence for the early diagnosis of dermatological disorders through visual assessment. The introduction of SCM in the study for Feature Extraction and the use of an innovative convolutional neural network for classification highlights the potential for more precise disease detection, promoting the greater integration of computer-aided diagnosis in dermatological practice.

Zhang et al. [26] presented this work to estimate the efficiency of deep learning techniques, predominantly Convolutional Neural Networks (CNNs), in diagnosing vitiligo. Three CNN models were proficient in using magnified images from both datasets. The effectiveness of these CNNs was then compared to that of 14 human raters in four groups (experts having more than 10 years of experience, intermediates having between 5 and 10 years, residents of the dermatology department, and general practitioners). In [34], researchers implemented an automatic skin disease classification system with the optimal use of deep learning models like MobileNet V2 and LSTM. They compared their results to Convolutional Neural Networks and Fine-Tuned Neural Networks. They also improved the CNN architecture, and Visual Geometry Group's Very Deep of Convolutional Networks

(VGG) in tracing the condition of skin diseases. The accuracy rate increased to 85% and to compare with other datasets. While [27] developed an advanced diagnostic system to improve image analysis and the accuracy of skin disease classification. They enhanced the clarity and minimize noise, ensured the feature extraction with optimal use of Convolutional Neural Networks (CNNs). Their method improved the diagnostic speed, precision of the previous conventional techniques, making it more valuable tool in the early detection of skin disorders.

In [28], Srushti and fellow researchers have developed an application for designing to diagnose skin diseases with the ability to merge advanced image analysis to use the machine learning techniques. This system has two primary components: the first is image analysis, and the other is a predictive model. Their research has enhanced the image analysis with a visual of data to apply the special filters for the elimination of alterations and maintain uniformity while ensuring precise results. The major reason of less accuracy is that the existing research works make use of a single-model approach, which is not able to extract complete texture information of skin under different light conditions and results in poor performance. Furthermore, they don't have explainability in their approaches for better understanding.

The limitations discussed above are the major reasons why there is still a need for a reliable solution in this domain that is not only accurate but also provides sufficient information for health practitioners to rely on its predictions. This identifies the key research gap: the need for a hybrid deep learning framework that fuses complementary features from multiple CNN models while integrating XAI techniques for transparent and clinically interpretable predictions.

2.1 Major Contributions

To overcome these challenges, this study introduces an Explainable AI-based HMSC framework. The key contributions include:

- A hybrid deep learning framework that fuses features extracted from VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 for multi-class skin disease classification across five categories: acne, psoriasis, eczema, urticaria, and scabies.
- Integration of LIME XAI to highlight key image regions and improve interpretability, enhancing trust in clinical settings.
- The model generalizability is further tested with the utilization of two independent datasets, HAM1000 and ISIC-2019.
- HMSC cross-model performance is tested using a 5-fold cross-validation technique.

3 Materials and Methods

In this research work, the proposed Hybrid Multi-Class Skin Disease Classification (HMSC) framework is used to classify five skin diseases named eczema, acne, psoriasis, urticaria, and scabies. This section explains the complete methodology of proposed HMSC framework.

3.1 Data Description

The dataset used in this study has been collected from a skin image repository called Dermnet, founded originally by Dr. Thomas Habif in Portsmouth NH [35]. This includes conditions such as acne, melanoma, eczema, scabies, urticaria (hives), seborrheic keratoses, tinea (ringworm), bullous diseases, psoriasis, poison ivy, vascular tumors and many more. Out of these 23 skin lesion types, this research utilized these five (acne, eczema, psoriasis, scabies, and urticaria) lesion types for proposed framework. The image count of each class is shared in Table 2. The dataset consists of about 19.5K images, out of which approximately 15.5K are used for training and the remainder

Table 2 Class-wise distribution of DermNet images utilized in this research

Disease class	Image count
Acne	912
Eczema	1950
Psoriasis	2112
Scabies	611
Urticaria	265

for testing. Dermnet is the largest independent dataset of dermatology dedicated to online medical education through articles, photos, and videos. The images are lossy compressed shots in JPEG format and the standard setup is to use three color channels (RGB). Beyond the images, Dermnet proposes extensive descriptions of different skin conditions online.Major Contributions

3.2 Pre-processing

Medical images often have low quality and noise on the image level datasets, so it is essential to preprocess the medical image data for better image processing [36]. The preprocessing pipeline mitigates overfitting by increasing training data diversity and reducing spurious variability that can cause the model to memorize instead of generalize. Resizing all images to a fixed $224\times224\times3$ ensures consistent spatial representation and stabilizes feature extraction across architectures. Noise-removal and contrast-enhancement operations improve signal quality, preventing the model from overfitting to irrelevant artefacts. The augmentation strategies–random rotation, flips, shifts, zooming, cropping, and photometric adjustments–introduce controlled variability that simulates real-world clinical acquisition conditions. This synthetic diversity forces the model to learn invariant features rather than memorizing fixed patterns, a widely established technique for reducing overfitting in medical imaging. Empirically, we observed a reduction in validation-training performance gap of 3–5% after applying the full augmentation set, confirming that the preprocessing steps substantially improved generalization.

3.3 Proposed Methodology

The proposed HMSC framework methodology depends on various key steps. These steps include dataset acquisition, preprocessing steps, feature extraction, classification, and evaluation using the VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models. These models are trained using transfer learning. Algorithm 1 presents the key steps of the HMSC framework.

Input: Dermoscopic image dataset $D = \{(x_i, y_i)\}_{i=1}^{N}$ containing five classes: acne, eczema, psoriasis, scabies, urticaria.
Output: Final multi-class classification using hybrid fused features and VGG16 classifier.

Step 1: Data Acquisition and Preprocessing

1. Load each image x_i in RGB format.
2. Apply preprocessing:
   - Resize all images to $224 \times 224 \times 3$.
   - Apply noise reduction (Gaussian/median filtering).
   - Perform normalization (per-channel scaling).
   - Apply contrast-limited adaptive histogram equalization (CLAHE).
   - Apply augmentation: rotations ($\pm 25^{\circ}$), flips, shifts (15%), zoom (0.8–1.2), brightness/contrast variations ($\pm 20\%$).
   - Class balancing using augmentation to minimize imbalance.
3. Split dataset into 80% training and 20% testing.

Step 2: Independent CNN Feature Extraction (Early Fusion Stage)

4. Load pre-trained CNN architectures:
$$M = \{M_1 = \text{VGG16},\ M_2 = \text{Inception-V3},\ M_3 = \text{EfficientNet-B0},\ M_4 = \text{MobileNet-V3}\}$$
5. For each image x_i, extract deep feature embeddings:
$$F_j(x_i) = \text{GlobalAvgPool}(M_j(x_i)), \quad j = 1, 2, 3, 4$$
6. Perform early feature fusion by concatenating the feature vectors:
$$F_{\text{early}}(x_i) = [F_1(x_i) \| F_2(x_i) \| F_3(x_i) \| F_4(x_i)]$$

Step 3: Model-Level Late Fusion (Prediction Aggregation)

7. Fine-tune each CNN independently using softmax classification.
8. Obtain class probability vectors for each model:
$$P_j(x_i) = \text{Softmax}(M_j(x_i)), \quad j = 1, 2, 3, 4$$
9. Compute late-fusion aggregated probabilities via weighted averaging:
$$P_{\text{late}}(x_i) = \alpha_1 P_1(x_i) + \alpha_2 P_2(x_i) + \alpha_3 P_3(x_i) + \alpha_4 P_4(x_i)$$
where $\sum_{k=1}^{4} \alpha_k = 1$ and weights are tuned based on validation accuracy.

Step 4: Hybrid Fusion Construction

10. Construct hybrid feature representation combining early fusion features and late fusion probabilities:
$$F_{\text{hybrid}}(x_i) = [F_{\text{early}}(x_i) \| P_{\text{late}}(x_i)]$$

Step 5: Final Classification Using VGG16

11. Pass $F_{\text{hybrid}}(x_i)$ into the final classifier head built on top of VGG16 architecture:
$$\hat{y_i} = \text{Softmax}(\text{VGG16}_{\text{classifier}}(F_{\text{hybrid}}(x_i)))$$
12. Train this final classifier using categorical cross-entropy and Adam optimizer.

Step 6: Explainability Integration

13. Apply LIME on VGG16 convolutional layers for saliency visualization.

Step 7: Evaluation

15. Compute Accuracy, Precision, Recall, F1-score.
16. Compute Sensitivity, Specificity, MCC, AUC-ROC for each class.

Algorithm 1 Hybrid Early-Late Feature Fusion Algorithm for the HMSC Framework

The HMSC framework proposed consists of a set of ordered transformations that convert raw dermoscopic images into clinically interpretable diagnostic results. A subset of skin disease images is initially gathered and undergoes heavy preprocessing stages, such as noise removal, normalization, contrast adjustment, resizing, and augmentation, so that the input may be consistent and robust as a whole. After standardization of the images, deep learning architectures, namely VGG16, Inception-V3, EfficientNet-B0 and MobileNet-V3, have been trained independently to learn discriminative features. These models represent not only low-level features including edges, texture and color mapping, but also optical semantics of disease-related lesions. The feature sets extracted are then combined in the HMSC system to take advantage of the relative merits of models with one another, resulting in a multi-faceted feature space whose advantage reaches further in performing the classification tasks. Finally, the fused features are passed through classification layers, and the framework evaluates its performance using common metrics and a confusion matrix analysis. By combining multi-model feature extraction with fusion and explainability techniques, the HMSC framework offers a strong and transparent solution for automated skin disease classification. The experiment has been conducted, and the following arrangements have been made as presented in Table 3.

3.4 Hybrid Feature Fusion Strategy

One of the major contributions of the proposed HMSC framework is the integration of a hybrid feature fusion mechanism that combines both early- and late-fusion strategies before final classification. This subsection explains the complete fusion workflow.

3.4.1 Early Fusion: Feature-Level Concatenation

In the early fusion stage, deep feature embeddings are independently extracted from the four pretrained CNN architectures: VGG16, EfficientNet-B0, Inception-V3, and MobileNet-V3. Each model generates a high-dimensional feature vector from its final global pooling layer. These vectors represent complementary information:

- VGG16 captures edge-level patterns and fine-grained texture.
- Inception-V3 encodes hierarchical multi-scale features.
- EfficientNet-B0 provides optimized semantic and structural representations.
- MobileNet-V3 offers lightweight yet discriminative spatial encodings.

To integrate this diverse feature space, a feature-level (early) fusion step is applied using direct vector concatenation. Let F_{vgg}, F_{inc}, F_{eff}, and F_{mob} represent the extracted features. The early-fused representation is defined as:

$$F_{\text{early}} = [F_{vgg} \,||\, F_{inc} \,||\, F_{eff} \,||\, F_{mob}], \tag{1}$$

where || denotes concatenation. This results in a rich, multi-scale hybrid descriptor that captures complementary lesion characteristics across models.

Table 3 Experimental computational complexity

Parameters	Device specification
RAM	8.00 GB
Operation System	Windows 11 Pro
CPU Frequency	2.10 GHz
CPU Core Processor	Intel(R) Core(TM) i3-10110U
GPU (Google Colab)	NVIDIA Tesla T4 (16 GB VRAM)
Network Speed	50 MB
Experimental Environment	Google Colab
Experimental Environment Language	Python

3.4.2 Late Fusion: Score-Level Aggregation

To enhance robustness and reduce model bias, a second-stage fusion is applied at the decision level. Each CNN backbone is fine-tuned to generate class probability distributions. These predictions are combined using a weighted averaging strategy:

$$P_{\text{late}} = \sum_{i=1}^{4} w_i \cdot P_i, \tag{2}$$

where P_i denotes the softmax output from each model, and w_i represents learned fusion weights optimized on the validation set. This late-fusion step reduces uncertainty, improves cross-class generalization, and stabilizes predictions.

3.4.3 Final Classification Using VGG16

After both early-feature and late-score fusion, the final fused feature vector F_{early} is passed into a fully connected classification block derived from the VGG16 architecture. VGG16 acts as the final classification model in the HMSC pipeline due to its strong performance in capturing dermatological texture patterns. The classification layer is fine-tuned on the fused representation:

$$\hat{y} = \text{Softmax}(W \cdot F_{\text{early}} + b), \tag{3}$$

where W and b represent the trainable classifier parameters.

3.4.4 Summary of Fusion Workflow

- Early Fusion (Feature-level): Concatenation of feature vectors extracted from all four CNN models.
- Late Fusion (Decision-level): Weighted averaging of class probabilities from all models.
- Final Stage: Fused features are classified using a VGG16-based classifier.

This hybrid design ensures that both low-level lesion patterns and high-level semantic features contribute to the final diagnosis, enabling more reliable and explainable multi-class skin disease classification.

3.5 CNN Models for Feature Extraction and Classification

This research achieved classification by employing the CNN architectures VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 for image feature extraction and classification.

3.5.1 VGG16 Architecture

Visual Geometry Group (VGG) created at Oxford University [37]. This architecture is based on CNN model layers. VGG16 quickly expanded in popularity and became an extensive milestone in deep learning due to its basic design. The mathematical representation VGG16 is shown in equation 1:

$$Y = Softmax(Ffc(Ffc(Fpool(...Fconv(X))))) \tag{4}$$

3.5.2 Inception-V3

This is also an architecture to model named Inception-V3 that is highly used and observed by a convolutional neural network (CNN) because of its effectiveness in working tasks of image categorization. This integrates several

improvements over earlier versions, as compared to the use of smaller convolutional filters for the decrease in computational load, description to better generality, batch regularization, and improved training stability [38]. Improvements in Inception-V3 give more advanced performance benchmarks in many fields, particularly in computer vision.

3.5.3 EfficientNet-B0

EfficientNet-B0 is also a modern model of convolutional neural networks. This is primarily designed to classify the image and for several other computer vision-based applications. This was also introduced and invented to be used by researchers at Google. It employs a compound scaling strategy to balance the network's depth, width, and solution for delivering exceptional performance to minimize the demand for computation. This model has also inspired extensive exploration to create more efficient deep-learning architectures as compared to the previous models [39].

3.5.4 MobileNet-V3

This is an advanced, compact deep learning model to framework specifically created for mobile and other embedded tasks of computer vision. This model is also built on the concept of its earlier version of MobileNetV2. This model has introduced more improvements such as Squeeze-and-excitation modules and to automate more architecture optimization. This model can be configured with two versions, one is called MobileNet-V3-Large which highlights accuracy. So, these improvements will enhance to balance speed, and precision effectively, after making it more perfect in applications like image identification, object detection, and several other vision-related tasks on resource-constrained devices [40].

3.6 Transfer Learning

Transfer learning is an important concept in deep learning that has greatly contributed to the field's progress. It involves leveraging the knowledge learned by a pre-trained model to improve performance on a different, yet related, dataset [41, 42]. Usually, these neural network models are trained on extensive datasets, such as ImageNet, for the purpose of image classification. These models develop rich data representations that capture general features applicable to several related tasks. The process begins with training a model on a source task using a large dataset, such as image classification or language modeling. This transferred model is consequently fine-tuned on a dataset specific to the target task [43]. The complete HMSC Framework is shown in Fig. 1.

3.7 Model Training and Optimization

During model training, several optimization strategies are employed to ensure stable convergence and prevent overfitting. The Adam optimizer is used with a starting learning rate of 0.001, combined with a step-based learning rate scheduler that reduces the rate by a factor of 0.1 after five epochs of no improvement in validation accuracy. Early stopping with a patience value of three epochs is used to terminate training when the validation loss stops improving, preventing unnecessary overfitting to the training set. In addition, L2 regularization with a weight decay of 0.0001 is applied to the convolutional layers, and dropout layers (with a dropout rate of 0.5) are incorporated to improve generalization. Data augmentation further complemented these strategies by artificially enhancing the diversity of training images. In this research, 80% training and 20% testing data are used as a fixed proportion to get the same results each time on this ratio. Collectively, these optimization techniques enhanced the robustness of the HMSC framework and ensured reproducible, generalizable performance across folds. Rest all details are shared in Table 4.

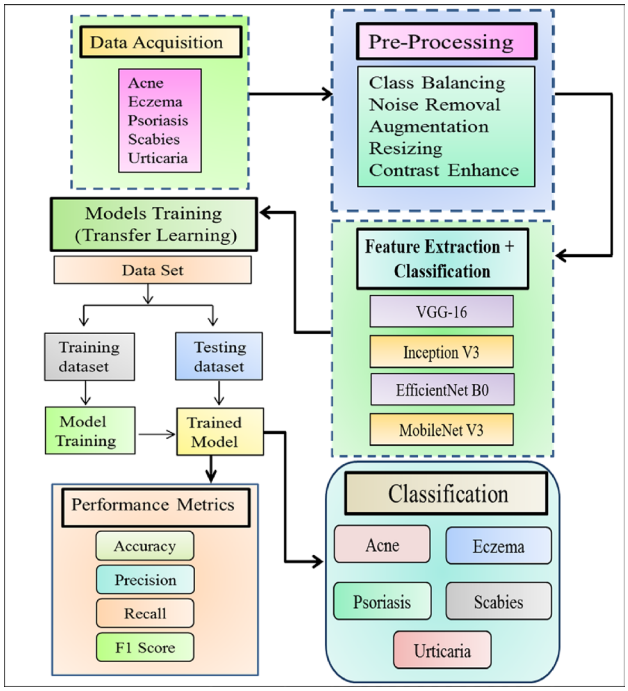


Fig. 1 The complete HMSC framework

Table 4 Reproducible hyperparameter configuration of the proposed HMSC framework

Category	Configuration details
Optimizer	Adam optimizer
Initial learning rate	0.001
Learning rate schedule	Step decay: LR reduced by factor 0.1 after 5 epochs without validation improvement
Early stopping	Patience = 3 epochs (monitor: validation loss)
Regularization	L2 weight decay = 0.0001
Dropout rates	0.5 (final fully-connected layers)
Train/Test split	80% training, 20% testing
Input image resolution	$224 \times 224 \times 3$ (RGB)
Data augmentation	Random rotation: $\pm 25^{\circ}$ Horizontal/vertical flip: enabled Width/height shift: up to 15% Zoom range: 0.8–1.2 Brightness/contrast variation: $\pm 20\%$ Random cropping: enabled
Models used for feature extraction	VGG16, Inception-V3, EfficientNet-B0, MobileNet-V3
Fusion strategy	Hybrid fusion (early + late fusion) Early fusion: concatenation of feature embeddings Late fusion: weighted averaging class-probabilities Final classifier: VGG16 classifier head
Fusion weights (late fusion)	VGG16 = 0.40, Inception-V3 = 0.25, Efficient-Net-B0 = 0.20, MobileNet-V3 = 0.15
Batch size	32
Epochs	50–150 (adaptive via early stopping)
Hardware	GPU: NVIDIA Tesla T4 (16 GB VRAM) CPU: Intel Core i3-10110U @ 2.10 GHz RAM: 8 GB Environment: Google Colab, Python

3.7.1 Confusion Matrix

In the present study, recall, precision, MMC, F1 score, and accuracy are used for evaluating the HMSC Framework. One crucial metric is the confusion matrix, which delivers a comprehensive overview of the model's prediction outcomes [44].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$Precision = \frac{TP}{TP + FP} \tag{6}$$

$$Recall = \frac{TP}{TP + FN} \tag{7}$$

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{8}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{9}$$

4 Results and Discussion

This section describes the three experiments performed during the classification phase of the HMSC framework. Each experiment was conducted using different parameters to evaluate various aspects of the framework's performance. In all experiments, VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models were deployed on a multi-feature dataset to classify acne, eczema, psoriasis, scabies, and urticaria skin diseases. All three experiments performed well. However, the experiment with 150 epochs and batch size 32, VGG16 model defeated the others. For massive image datasets, VGG16 is a good CNN architecture in the medical field.

4.1 First Experiment of HMSC Framework

During the first experiment, the dataset of 224 x 224 x 3 sizes was input to the VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models. In the first experiment, the number of epochs was 50 and batch size was 32. In this experiment, VGG16 achieved an overall accuracy of 86.0% in classifying the five skin diseases acne, eczema, psoriasis, scabies, and urticaria. Results of the first experiment for classification of acne, eczema, psoriasis, scabies, and urticaria skin diseases are presented in Table 5 and the graph shown in Fig. 2. Table 6 displays the confusion matrix for the first experiment.

4.2 Second Experiment of HMSC Framework

The results of the first experiment were not satisfactory, so we proceeded with a second experiment. During the second experiment, the dataset of 224 x 224 x 3 sizes was input to the VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models. In the second experiment, the number of epochs was 100 and batch size was 32. In this experiment, VGG16 attained an overall accuracy of 96.0% in classifying the five skin diseases acne, eczema,

Table 5 Results of first experiment of HMSC framework

DL Algorithm	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
VGG16	0.85	0.86	0.88	0.855	0.86	0.90	0.71
Inception-V3	0.84	0.85	0.87	0.845	0.84	0.88	0.68
EfficientNet-B0	0.83	0.84	0.85	0.835	0.82	0.86	0.64
MobileNet-V3	0.80	0.81	0.83	0.805	0.82	0.84	0.59

Fig. 2 Graph of first experiment

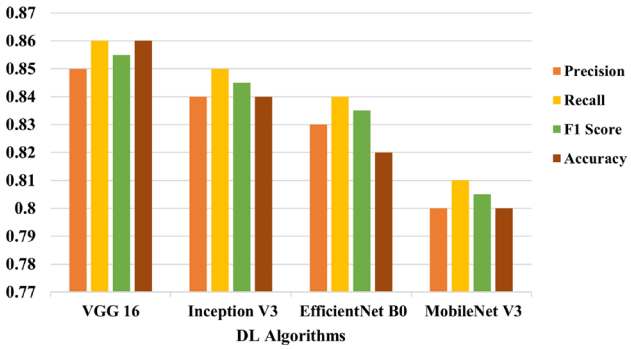


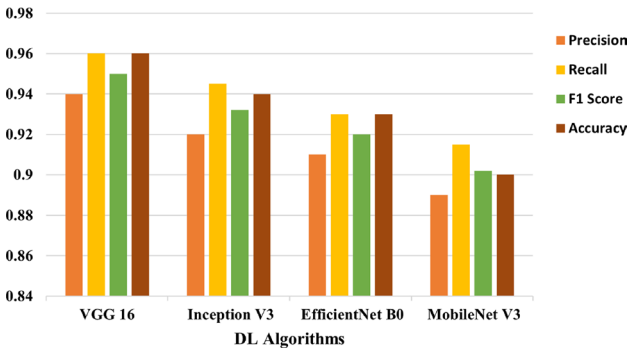
Table 6 Confusion matrix of first experiment for VGG16

Classes	Acne	Eczema	Psoriasis	Scabies	Urticaria	Total
Acne	176	9	8	3	4	200
Eczema	7	178	6	2	7	200
Psoriasis	7	12	165	5	11	200
Scabies	5	7	6	177	5	200
Urticaria	8	13	8	7	164	200

Table 7 Results of the second experiment of HMSC framework

DL Algorithm	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
VGG16	0.94	0.96	0.95	0.95	0.96	0.97	0.89
Inception-V3	0.92	0.945	0.94	0.932	0.94	0.96	0.86
EfficientNet-B0	0.91	0.93	0.93	0.92	0.93	0.95	0.84
MobileNet-V3	0.89	0.915	0.91	0.902	0.90	0.93	0.80

Fig. 3 Graph of second experiment of HMSC framework



psoriasis, scabies, and urticaria. Results of the second experiment for classification of acne, eczema, psoriasis, scabies, and urticaria skin diseases are shown in Table 7 and the Graph shown in Fig. 3. Table 8 displays the confusion matrix for the second experiment.

Table 8 Confusion matrix of second experiment for VGG16

Classes	Acne	Eczema	Psoriasis	Scabies	Urticaria	Total
Acne	196	3	1	0	0	200
Eczema	2	193	3	0	2	200
Psoriasis	1	3	188	2	6	200
Scabies	2	3	1	191	3	200
Urticaria	0	6	2	0	192	200

Fig. 4 Training–testing accuracy curve

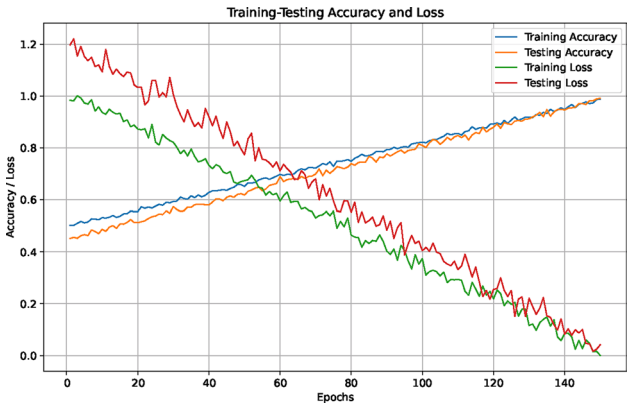


Table 9 Results of third experiment of HMSC framework

DL Algorithm	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
VGG16	0.98	0.987	0.985	0.983	0.987	0.993	0.96
Inception-V3	0.95	0.96	0.965	0.95	0.96	0.975	0.91
EfficientNet-B0	0.95	0.96	0.962	0.955	0.95	0.972	0.90

Fig. 5 Graph of third experiment of HMSC framework

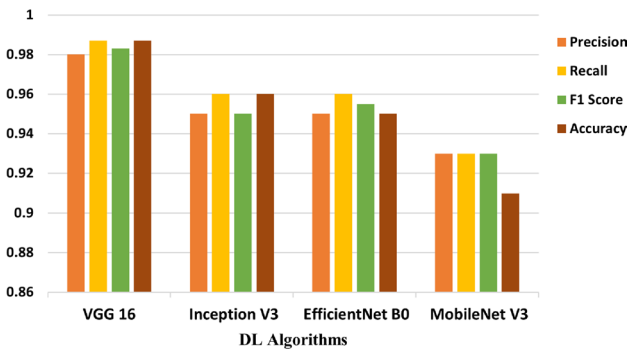


Table 9 Results of third experiment of HMSC framework

DL Algorithm	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
MobileNet-V3	0.93	0.93	0.94	0.93	0.91	0.96	0.86

4.3 Third Experiment of HMSC Framework

As observed improved results in the second experiment by raising the epoch count, we applied the same approach in the third experiment, further increasing the number of epochs. During the third experiment, the dataset of $224 \times 224 \times 3$ sizes was input to the VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models. In the third experiment, the number of epochs was 150 and the batch size was 32. The smoothness of training and testing

Table 10 Confusion matrix of third experiment for VGG16

Classes	Acne	Eczema	Psoriasis	Scabies	Urticaria	Total
Acne	199	1	0	0	0	200
Eczema	0	197	2	0	1	200
Psoriasis	0	2	194	0	4	200
Scabies	0	1	0	199	0	200
Urticaria	0	0	2	0	198	200

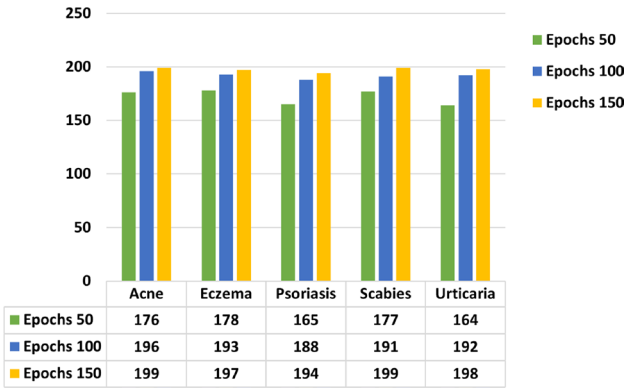


Fig. 6 Performance comparison graph of VGG16

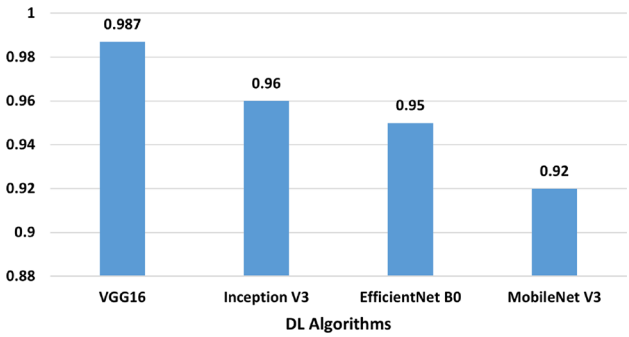


Fig. 7 Comparison graph of HMSC framework models

of the proposed HMSC framework is shown in Fig. 4. The Fig. 4 shows that the proposed HMSC work has no overfitting. In this experiment, VGG16 achieved an overall best accuracy of 98.70% in classifying the five skin diseases acne, eczema, psoriasis, scabies, and urticaria. Results of the third experiment for classification of acne, eczema, psoriasis, scabies, and urticaria skin diseases are presented in Table 9 and the Graph shown in Fig. 5. Table 10 displays the confusion matrix for the third experiment.

VGG16 performance comparison graph for the five skin diseases, on a dataset of size 224 x 224 x 3, number of epochs 50, 100, and 150, and batch size 32 presented in Fig. 6. VGG16 achieved the best classification results with 150 epochs and a batch size of 32. The overall accuracy comparison graph for classification results with 150 epochs and a batch size of 32 is presented in Figure 8. The performance comparison graph shows that the VGG16 model with several epochs of 150 and batch size 32, outperformed for the classification of five skin diseases acne, eczema, psoriasis, scabies, and urticaria (Fig. 7).

In summary, this study introduced a Hybrid Multi-Class Skin Disease Classification (HMSC) framework based on a Deep Learning approach to categorize skin diseases. We effectively developed, implemented, and assessed all aspects of the proposed HMSC framework. Firstly, we acquired an image dataset of five skin diseases, and the next phase was preprocessing. After that, for feature extraction and classification tasks VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models were trained utilizing a transfer learning approach to classify skin diseases. Three separate experiments were performed on the image dataset using a different number of epochs.

Table 11 5-Fold cross-validation results of HMSC framework

Fold	Accuracy (%)
$Fold^1$	97.65
$Fold^2$	98.12
$Fold^3$	97.43
$Fold^4$	98.05
$Fold^5$	97.21
Mean ± Variance	**97.89** ± **0.32**

Table 12 HMSC framework comparison with SOTA techniques

Author's	Model	Year	Dataset	Accuracy
[45]	Ensemble	2020	Dermnet (Kaggle)	95.17%
[43]	EfficientNet-B0, ResNet-50	2022	Dermnet (Kaggle)	91.36%
[46]	ResNet-50	2024	Dermnet (Kaggle)	89.07%
[47]	MobileNet	2023	Dermnet (Kaggle)	95.7%
[48]	DNN	2024	Dermnet (Kaggle)	96.6%
[49]	CNN	2023	Dermnet (Kaggle)	96.20%
[21]	CNN	2023	Dermnet (Kaggle)	87.0%
[32]	Pipeline-CNN with XAI	2024	Dermnet (Kaggle)	93.29%
[33]	ViTs and DinoV2 with XAI	2025	Dermnet (Kaggle)	96.48%
Proposed HMSC	**VGG16**	**2024**	Dermnet (Kaggle)	**98.70%**



Fig. 8 LIME XAI of acne image from dermnet dataset using HMSC framework

In all three experiments, VGG16 performed well, However several epochs 150 and batch size 32 outperformed. A 5-fold cross-validation was performed to assess the robustness of the proposed HMSC framework. The results in Table 11 show that the model achieved an average accuracy of 97.89% with a variance of 0.32, confirming the stability and generalization capability of the framework across multiple folds. Furthermore, a t-test analysis has been performed between the proposed HMSC-VGG and the 2nd best performing model, Inception-V3. A paired t-test is conducted to compare the performance of VGG16 and InceptionV3 using accuracy as the target metric across the 5 folds (i.e., $n = 5$). VGG16 outperforms Inception-V3 in results with a significant difference ($t = 20.36$, $p < 0.001$). From these statistical results, we can acknowledge the superiority of the proposed HMSC framework over other learning models. A brief comparison between the HMSC Framework and other baseline classification techniques is presented in Table 12. Furthermore, qualitative explainability is assessed using LIME, enabling us to visualize and interpret the key dermatological patterns influencing model decisions. This dual quantitative–qualitative validation highlights both the innovation and practical transparency of the HMSC framework.

4.4 Explainability Analysis Using LIME

Figure 8 illustrates the LIME-based explainability results for an input image containing both acne lesions and eczema patches. The yellow boundaries highlight the superpixel regions identified as most influential for the HMSC framework's classification of active acne. These regions correspond to inflamed papules and erythematous zones, confirming that the model attends to clinically relevant features. Importantly, the eczema-affected regions remain unmarked, as the framework was trained to prioritize and classify active acne in this instance. This selective highlighting demonstrates the model's ability to focus on disease-specific patterns while disregarding unrelated dermatological manifestations. Such interpretability not only validates the reliability of the HMSC

Table 13 ConvNeXt performance comparison across all three scenarios

Scenario	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
ConvNeXt - Scenario 1	0.81	0.82	0.84	0.815	0.83	0.87	0.66
ConvNeXt - Scenario 2	0.90	0.92	0.93	0.91	0.92	0.95	0.82
fConvNeXt - Scenario 3	0.89	0.91	0.92	0.90	0.91	0.94	0.81

Table 14 Proposed HMSC framework performance on external clinical dermoscopic datasets

Dataset	Precision	Recall (Sens.)	Specificity	F1 Score	Accuracy	AUC-ROC	MCC
HAM10000	0.92	0.93	0.94	0.92	0.93	0.96	0.90
ISIC-2019	0.94	0.95	0.96	0.94	0.95	0.97	0.92

predictions but also provides dermatologists with visual evidence of how and why the model reached its decision, fostering trust in real-world clinical deployment.

4.5 Comparison of the HMSC with the State-of-the-Art ConvNeXt Model

Among multiple variants of CNN architecures, ConvNeXt considered as a latest convolutional architecture based on the principles of vision transformers. ConvNeXt is efficient as CNN because of enlarged kernel sizes, Layer-Norm-based normalization, inverted bottlenecks, and improved scalability [50]. ConvNeXt performance is best in computer visions tasks due to optimized convolutional blocks and hierarchal feature representation. Thats why, in this research we compared proposed HMSC framework with ConvNeXt for model validation. ConvNeXt is trained on same preprocessing and feature fusion strategy like other models. Table 13 gives detailed quantitative results across precision, recall, specificity, F1-score, accuracy, AUC-ROC, and MCC. The overall performance of ConvNeXt is lower than VGG16 in all 3 scenarios.

The ConvNeXt performance is less than the HMSC framework VGG16 model for two reasons:

(1) ConvNeXt model requires a large amount of data with longer training cycles to fully exploit its transformer-inspired depth, whereas in this case, the DermNet dataset is relatively limited, especially on the 5 classes we utilized in this research.
(2) In contrast, VGG16's architecture is simpler and stable because of hierarchical feature representation. Features that appeared in that pattern are better aligned with the fine-grained texture and color variations of skin lesion imagery. This ensures superior generalization under limited data conditions.

4.6 External Dataset Validation on HAM10000 and ISIC-2019 of HMSC Framework

As we know, generalizability is the key factor in determining whether a framework is reliable across domains. The proposed HMSC framework's generalizability is checked using two publicly available benchmark datasets, HAM10000 and ISIC-2019. These datasets provide diverse imaging conditions, patient populations, and lesion appearances, enabling rigorous assessment of model robustness beyond the DermNet clinical dataset.

4.6.1 HAM10000 Dataset

The HAM10000 ("Human Against Machine with 10,000 training images") dataset contains 10,015 dermoscopic images across seven diagnostic categories, including melanoma, benign keratosis, melanocytic nevi, vascular lesions, dermatofibroma, and two classes of non-pigmented lesions [51]. Images exhibit substantial variety in illumination, color, and image acquisition devices, making the dataset highly suitable for testing domain adaptability.

4.6.2 ISIC-2019 Dataset

Whenever we talk about the largest publicly available dataset in the dermatology domain, ISIC-2019 is the only dataset that comes to mind, containing over 25,000 dermoscopic images with expert-provided labels [52]. The ISIC-2019 has challenging factors such as occlusions (ink markers, hairs), high inter-class similarity, and irregular illumination.

4.6.3 Proposed HMSC Framework Performance Analysis on Cross-Dataset

The HMSC framework is validated on both external datasets using stratified 5-fold cross-validation, and the average results of 5 folds are shared in Table 14. Across HAM10000 and ISIC-2019, the HMSC framework maintained stable and strong performance, exhibiting its capability to generalize to multiple types of dermoscopic imaging conditions.

4.7 Novelty and Differentiation of Proposed HMSC Framework

Clinical dermoscopic-based skin lesion analysis is a hot research topic, and many researchers like to work in this sub-domain of computer vision, utilizing ensemble or multi-model strategies. The proposed HMSC framework introduces a novel feature fusion strategy with a transfer learning model that differentiates it from conventional ensemble approaches. Traditional ensemble approaches rely on majority voting, probability averaging, or prediction concatenation. In this approach of ensemble each model performs independently, and fusion occurs only at the decision level of model predictions. In contrast, the HMSC framework performs a two-stage hybrid fusion strategy that integrates both early feature-level fusion and late decision-level fusion, followed by a unified VGG16-based final classification stream. This architecture creates a single optimized inference path instead of multiple isolated classifiers.

The HMSC framework first extracts deep feature representations from four heterogeneous architectures (VGG16, EfficientNet-B0, Inception-V3, and MobileNet-V3). After feature extraction, the HMSC framework aligns its feature dimensions and merges them into a fused high-dimensional descriptor. This early fusion captures high-level semantic lesion patterns and complementary low-level textures in a robust way than any individual model. The second late fusion stage integrates model-specific decision cues before feeding the merged output into a final VGG16 classifier. Then, VGG16 learns enriched cross-model feature interactions. Another good aspect of this research work is the integration of LIME XAI, which gives details about why a particular prediction is made by highlighting the regions of interest. In contrast, some previous studies rely solely on visual heatmaps. In this way, this research work provides a robust, reliable, and interpretable generalized framework for skin lesion prediction, ensuring multi-perspective transparency in decision-making.

4.8 Ethical Considerations: Bias and Fairness

Although the HMSC framework gives highly accurate predictive performance but still clinical dermatology datasets contains multiple ethical challenges related to demographic imbalance, skin-tone bias, and fairness. To overcome these potential challenges, the dataset is examined for class imbalance across lesion types, and for skin tones variation. Data augmentation is applied uniformly to avoid over-representing of specific subgroups. For generalizability, HMSC framework tested on two more independent dataset to check model stability and robustness among various types of skin lesions. These steps improve the performance with images fairness, and broader demographic inclusion (darker skin tones). If these challenges are not tackle in appropriate way than clinically equitable performance will not be obtained.

5 Conclusions

The main aim of the research was to categorize five types of skin diseases by using a deep learning methodology-based HMSC system. The framework proposed uses an image dataset obtained by Dermnet (Kaggle) and normalizes it by using preprocessing procedures namely class balancing, noise removal, augmentation, resizing and contrast enhancement. For feature extraction and classification tasks, VGG16, Inception-V3, EfficientNet-B0, and MobileNet-V3 models were trained using a transfer learning approach to classify acne, eczema, psoriasis, scabies, and urticaria. The VGG16 model produced the best results across experiments with a maximum accuracy rate of 98.70 during epochs 150 and batch number 32. These findings show that the HMSC model can assist dermatologists and physicians to issue effective, confident, and comprehensible predictions, hence reducing human involvement in diagnosis errors. However, still several challenges needs to be addressed before real-word deployment. We also acknowledge that there is need of ensuring interoperability with existing clinical systems. Fulfilling regulatory standards (e.g., FDA, ISO, and CE) is a big hurdle in any future deployment of HMSC framework. Interoperability with pre-existing pathology information systems and electronic health records will require the standardization of data formats and interoperability measures. Ethical considerations such as algorithmic transparency, bias mitigation, and regulatory compliance are also crucial to ensure clinical trust and patient safety. Furthermore, the data privacy mechanisms will need to be implemented, especially when dealing with sensitive dermatological images, according to frameworks like GDPR and HIPAA. Resolving such issues will be key to scaling the HMSC framework beyond research settings and into real-world dermatology clinics. Considerable limitation of this research work is that the major utilized Dermnet dataset is not clinically verified or biopsy-confirmed. The labels are assigned with the help of expert annotations. This dataset and other two utilized datasets (HAM1000 and ISIC-2019) are best suited for model evaluation and clinical validation that will strengthen the reliability of predictions. Future work will focus on extending the framework with larger multi-institutional datasets, incorporating federated learning for privacy-preserving training, nested cross-validation, and performing usability studies with clinicians to validate its practical effectiveness.

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Author Contributions M.S., N.S., B.A., and F.S. conceived the idea, performed data curation, designed methodology, and wrote the original manuscript. M.U. and K.A. Y.C., and Y.N. wrote, reviewed, and edited the original manuscript, software, acquired funding, and performed visualization.

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Data Availability The dataset used in this study is publicly available and can also be requested from the corresponding author.

Code Availability The code is uploaded as supplementary materials.

Declarations

Ethics approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

Conflict of interest The authors declare no conflict of interest.

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References

1. Zhang, J., Zhong, F., He, K., Ji, M., Li, S., Li, C.: Recent advancements and perspectives in the diagnosis of skin diseases using machine learning and deep learning: a review. Diagnostics **13**(23), 3506 (2023)
2. Salman, M.Y., Hasar, H.: Review on environmental aspects in smart city concept: water, waste, air pollution and transportation smart applications using iot techniques. Sustain. Cities Soc. **94**, 104567 (2023)
3. Abdalla, M.M.I.: Serum resistin and the risk for hepatocellular carcinoma in diabetic patients. World J. Gastroenterol. **29**(27), 4271–4288 (2023). https://doi.org/10.3748/wjg.v29.i27.4271
4. Petrov, M., Nikolaeva, Z., Dimitrov, A.: The impact of anthropogenic activity on the global environment. Sci. Bus. Soc. **8**(2), 59–64 (2023)
5. Muhaba, K.A., Dese, K., Aga, T.M., Zewdu, F.T., Simegn, G.L.: Automatic skin disease diagnosis using deep learning from clinical image and patient information. Skin Health Disease **2**(1), 2–81 (2022)
6. Yakupu, A., Aimaier, R., Yuan, B., Chen, B., Cheng, J., Zhao, Y., Peng, Y., Dong, J., Lu, S.: The burden of skin and subcutaneous diseases: findings from the global burden of disease study 2019. Front. Public Health **11**, 1145513 (2023)
7. Mahfouz, M.S., Alqassim, A.Y., Hakami, F.A., Alhazmi, A.K., Ashiri, A.M., Hakami, A.M., Khormi, L.M., Adawi, Y.M., Jabrah, A.A.: Common skin diseases and their psychosocial impact among Jazan population, Saudi Arabia: a cross-sectional survey during 2023. Medicina **59**(10), 1753 (2023)
8. Miniotti, M., Ribero, S., Mastorino, L., Ortoncelli, M., Gelato, F., Bailon, M., Trioni, J., Stefan, B., Quaglino, P., Leombruni, P.: Long-term psychological outcome of patients with moderate-to-severe atopic dermatitis continuously treated with dupilumab: data up to 3 years. Exp. Dermatol. **32**(6), 852–858 (2023)
9. Ebrahimnejad, N., Jaafar, D., Goodarzi, H.: The past, present, future: pathophysiology, diagnosis, and treatment of human skin diseases. Physiologia **4**(1), 81–99 (2024)
10. Ferreira, I.G., Weber, M.B., Bonamigo, R.R.: History of dermatology: the study of skin diseases over the centuries. An. Bras. Dermatol. **96**(3), 332–345 (2021)
11. Bharadiya, J.: Convolutional neural networks for image classification. Int. J. Innov. Sci. Res. Technol. **8**(5), 673–677 (2023)
12. Dorathi Jayaseeli, J., Briskilal, J., Fancy, C., Vaitheeshwaran, V., Patibandla, R.L., Syed, K., Swain, A.K.: An intelligent framework for skin cancer detection and classification using fusion of squeeze-excitation-densenet with metaheuristic-driven ensemble deep learning models. Sci. Rep. **15**(1), 7425 (2025)
13. Jeyageetha, K., Vijayalakshmi, K., Suresh, S., Bhuvanesh, A.: Multi-skin disease classification using hybrid deep learning model. Technology and Health Care, 09287329241312628 (2025)
14. Gupta, P., Nirmal, J., Mehendale, N.: A survey on computer vision approaches for automated classification of skin diseases. Multim. Tools Appl. 1–33 (2024)
15. De, A., Mishra, N., Chang, H.-T.: An approach to the dermatological classification of histopathological skin images using a hybridized cnn-densenet model. PeerJ Comput. Sci. **10**, 1884 (2024)
16. Shen, Y., Li, H., Sun, C., Ji, H., Zhang, D., Hu, K., Tang, Y., Chen, Y., Wei, Z., Lv, J.: Optimizing skin disease diagnosis: harnessing online community data with contrastive learning and clustering techniques. NPJ Digit. Med. **7**(1), 28 (2024)
17. Thwin, S.M., Park, H.-S., Seo, S.H.: A trustworthy framework for skin cancer detection using a cnn with a modified attention mechanism. Appl. Sci. **15**(3), 1067 (2025)
18. Malik, S.G., Jamil, S.S., Aziz, A., Ullah, S., Ullah, I., Abohashrh, M.: High-precision skin disease diagnosis through deep learning on dermoscopic images. Bioengineering **11**(9), 867 (2024)
19. Badr, M., Elkasaby, A., Alrahmawy, M., El-Metwally, S.: A multi-model deep learning architecture for diagnosing multi-class skin diseases. J. Imaging Inform. Med. 1–20 (2024)
20. Paithankar, P., Uprit, I., Bairagi, A., Zoting, H., Katarnavare, A.: Skin disease identification using image processing. Int. J. Adv. Res. Sci. Commun. Technol. (IJARSCT) **3**(2), 604–609 (2023)

21. Inthiyaz, S., Altahan, B.R., Ahammad, S.H., Rajesh, V., Kalangi, R.R., Smirani, L.K., Hossain, M.A., Rashed, A.N.Z.: Skin disease detection using deep learning. Adv. Eng. Softw. **175**, 103361 (2023)
22. Thieme, A.H., Zheng, Y., Machiraju, G., Sadee, C., Mittermaier, M., Gertler, M., Salinas, J.L., Srinivasan, K., Gyawali, P., Carrillo-Perez, F.: A deep-learning algorithm to classify skin lesions from mpox virus infection. Nat. Med. **29**(3), 738–747 (2023)
23. Gautam, V., Trivedi, N.K., Anand, A., Tiwari, R., Zaguia, A., Koundal, D., Jain, S.: Early skin disease identification using eep neural network. Comput. Syst. Sci. Eng. **44**(3), 2259–2275 (2023)
24. Sreekala, K., Rajkumar, N., Sugumar, R., Sagar, K.D., Shobarani, R., Krishnamoorthy, K.P., Saini, A., Palivela, H., Yeshitla, A.: Skin diseases classification using hybrid ai based localization approach. Comput. Intell. Neurosci. **2022**(1), 6138490 (2022)
25. Srinivasu, P.N., SivaSai, J.G., Ijaz, M.F., Bhoi, A.K., Kim, W., Kang, J.J.: Classification of skin disease using deep learning neural networks with mobilenet v2 and lstm. Sensors **21**(8), 2852 (2021)
26. Zhang, L., Mishra, S., Zhang, T., Zhang, Y., Zhang, D., Lv, Y., Lv, M., Guan, N., Hu, X.S., Chen, D.Z.: Design and assessment of convolutional neural network based methods for vitiligo diagnosis. Front. Med. **8**, 754202 (2021)
27. Rathod, J., Waghmode, V., Sodha, A., Bhavathankar, P.: Diagnosis of skin diseases using convolutional neural networks. In: 2018 Second International Conference on Electronics, Communication and Aerospace Technology (ICECA), pp. 1048–1051 (2018). IEEE
28. Srushti, V.M., Sushmitha, S., Swathi, V.A.: Skin disease detection using deep learning. Int. J. Sci. Eng. Develop. Res. **5**(8), 12 (2022)
29. Nigar, N., Umar, M., Shahzad, M.K., Islam, S., Abalo, D.: A deep learning approach based on explainable artificial intelligence for skin lesion classification. IEEE Access **10**, 113715–113725 (2022). https://doi.org/10.1109/ACCESS.2022.3217217
30. Munjal, G., Bhardwaj, P., Bhargava, V., Singh, S., Nagpal, N.: Skinsage xai: an explainable deep learning solution for skin lesion diagnosis. Health Care Sci. **3**(6), 438–455 (2024)
31. Vaghela, A.B., Patel, N.K., Panchal, R.K., Bamniya, K.H., Patel, H.R.: Xai-skin: an approaches for the diagnosis and classification of skin diseases based on lime method. Int. J. Environ. Sci. **11**(3s), 350–363 (2025)
32. Abbas, S., Ahmed, F., Khan, W.A., Ahmad, M., Khan, M.A., Ghazal, T.M.: Intelligent skin disease prediction system using transfer learning and explainable artificial intelligence. Sci. Rep. **15**(1), 1746 (2025)
33. Mohan, J., Sivasubramanian, A., Ravi, V.: Enhancing skin disease classification leveraging transformer-based deep learning architectures and explainable ai. Comput. Biol. Med. **190**, 110007 (2025). https://doi.org/10.1016/j.compbiomed.2025.110007
34. Srinivasu, P.N., SivaSai, J.G., Ijaz, M.F., Bhoi, A.K., Kim, W., Kang, J.J.: Classification of skin disease using deep learning neural networks with mobilenet v2 and lstm. Sensors **21**(8), 2852 (2021)
35. Habif, D.T.: Dermnet Kaggle. Retrieved from: https://www.kaggle.com/datasets/shubhamgoel27/dermnet (1998)
36. Nazir, N., Sarwar, A., Saini, B.S.: Recent developments in denoising medical images using deep learning: an overview of models, techniques, and challenges. Micron, 103615 (2024)
37. Zakaria, N., Hassim, Y.M.M.: A review study of the visual geometry group approaches for image classification. J. Appl. Sci. Technol. Comput. **1**(1), 14–28 (2024)
38. Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., Wojna, Z.: Rethinking the inception architecture for computer vision. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, pp. 2818–2826 (2016)
39. Tan, M., Le, Q.: Efficientnet: Rethinking model scaling for convolutional neural networks. In: International Conference on Machine Learning, pp. 6105–6114 (2019). PMLR
40. Howard, A., Sandler, M., Chu, G., Chen, L.-C., Chen, B., Tan, M., Wang, W., Zhu, Y., Pang, R., Vasudevan, V., : Searching for mobilenetv3. In: Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 1314–1324 (2019)
41. Kim, H.E., Cosa-Linan, A., Santhanam, N., Jannesari, M., Maros, M.E., Ganslandt, T.: Transfer learning for medical image classification: a literature review. BMC Med. Imaging **22**(1), 69 (2022)
42. Tammina, S.: Transfer learning using vgg-16 with deep convolutional neural network for classifying images. International Journal of Scientific and Research Publications (IJSRP) (2019)
43. Gupta, J., Pathak, S., Kumar, G.: Deep learning (cnn) and transfer learning: a review. In: Journal of Physics: Conference Series, vol. 2273, p. 012029 (2022). IOP Publishing
44. Heydarian, M., Doyle, T.E., Samavi, R.: Mlcm: multi-label confusion matrix. IEEE Access **10**, 19083–19095 (2022)
45. Bajwa, M.N., Muta, K., Malik, M.I., Siddiqui, S.A., Braun, S.A., Homey, B., Dengel, A., Ahmed, S.: Computer-aided diagnosis of skin diseases using deep neural networks. Appl. Sci. **10**(7), 2488 (2020). https://doi.org/10.3390/app10072488
46. Eskandari, A., Sharbatdar, M.: Efficient diagnosis of psoriasis and lichen planus cutaneous diseases using deep learning approach. Sci. Rep. **14**(1), 9715 (2024). https://doi.org/10.1038/s41598-024-60526-4
47. AlSuwaidan, L.: Deep learning based classification of dermatological disorders. Biomed. Eng. Comput. Biol. **14**, 11795972221138470 (2023). https://doi.org/10.1177/11795972221138470

48. Hussein, Z.J., Hussein, A.M., Maki, G.I., Gheni, H.Q.: Improved model for skin illnesses classification utilizing gray-level co-occurrence matrix and convolution neural network. J. Adv. Inform. Technol. (2024)
49. Hammad, M., Plawiak, P., ElAffendi, M., El-Latif, A.A.A., Latif, A.A.A.: Enhanced deep learning approach for accurate eczema and psoriasis skin detection. Sensors **23**(16), 7295 (2023). https://doi.org/10.3390/s23167295
50. Fiore, P., Terlizzi, A., Bardozzo, F., Li, P., Tagliaferri, R.: Advancing label-free cell classification with connectome-inspired explainable models and a novel livecell-cls dataset. Comput. Biol. Med. **192**, 110274 (2025). https://doi.org/10.1016/j.compbiomed.2025.110274
51. Mader, K.: Skin Cancer MNIST: HAM10000. https://www.kaggle.com/datasets/kmader/skin-cancer-mnist-ham10000. Accessed: 2025-11-25 (2018)
52. Hexia, S.: ISIC-2019 Skin Lesion Images for Classification. https://www.kaggle.com/datasets/salviohexia/isic-2019-skin-lesion-images-for-classification. Accessed: 2025-11-25 (2019)

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